

Picard Iteration for the Characteristic Initial Value Problem in Einstein Equations

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Abstract

We present a first-order Picard iteration for vacuum and Einstein–scalar-field equations in double-null gauge. The numerical realization combines characteristic constraint solves, LGL spectral elements, pole-free spherical operators, Galerkin projection, and independent geometric residual audits.

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1 Introduction

Characteristic evolution exposes the transport structure of the Einstein equations while retaining a genuinely geometric angular problem. This article fixes the conventions, derives the iteration hierarchy, and states the discretization and residual definitions used by the accompanying numerical experiments.

2 Equations and Theoretical Analysis

2.1 Double null foliation and equations

In this section, we introduce the equations for Lorentzian metric under double null foliation. Let $(\mathcal{M}, g)$ be a 4-dimensional Lorentzian manifold. With double null coordinates (u, v, θ^A) , the metric can be written as

$$g = -2\Omega^2(du \otimes dv + dv \otimes du) + g_{AB}(d\theta^A - b^A du) \otimes (d\theta^B - b^B du), \quad (2.1)$$

and we use (g_{AB}, b^A, Ω) to represent the metric. The level sets of u and v are denoted by H_u and $\underline{H}_v$ respectively, and their spherical intersection $H_u \cap \underline{H}_v$ is denoted by $S_{u,v}$. With double null frame

$$e_3 = \Omega^{-1}(\partial_u + b), \quad e_4 = \Omega^{-1}\partial_v, \quad e_A = \partial_{\theta^A},$$

we can define Ricci coefficients, or connection components

$$\begin{aligned} \chi_{AB} &= g(D_A e_4, e_B), & \underline{\chi}_{AB} &= g(D_A e_3, e_B), \\ \eta_A &= -\frac{1}{2}g(D_3 e_A, e_4), & \underline{\eta}_A &= -\frac{1}{2}g(D_4 e_A, e_3), \\ \omega &= -\frac{1}{4}g(D_4 e_3, e_4), & \underline{\omega} &= -\frac{1}{4}g(D_3 e_4, e_3), \\ \zeta_A &= \frac{1}{2}g(D_A e_4, e_3) = -\frac{1}{4}\Omega^{-2}(\mathcal{L}_v b^B)g_{AB}. \end{aligned} \quad (2.2)$$

Their transport equations are listed below:

$$\begin{aligned} \nabla_4 \text{tr } \chi + \frac{1}{2}(\text{tr } \chi)^2 &= -\text{Ric}_{44} - |\hat{\chi}|^2 - 2\omega \text{tr } \chi, \\ \nabla_3 \text{tr } \underline{\chi} + \frac{1}{2}(\text{tr } \underline{\chi})^2 &= -\text{Ric}_{33} - |\hat{\underline{\chi}}|^2 - 2\underline{\omega} \text{tr } \underline{\chi}, \end{aligned} \quad (2.3)$$

$$\begin{aligned} \nabla_3 \hat{\chi}_{AB} + \frac{1}{2} \text{tr } \underline{\chi} \hat{\chi}_{AB} &= \widehat{\text{Ric}}_{AB} + 2\underline{\omega} \hat{\chi}_{AB} + (\nabla \hat{\otimes} \eta)_{AB} + (\eta \hat{\otimes} \eta)_{AB} - \frac{1}{2} \text{tr } \chi \hat{\chi}_{AB}, \\ \nabla_4 \hat{\underline{\chi}}_{AB} + \frac{1}{2} \text{tr } \chi \hat{\underline{\chi}}_{AB} &= \widehat{\text{Ric}}_{AB} + 2\omega \hat{\underline{\chi}}_{AB} + (\nabla \hat{\otimes} \underline{\eta})_{AB} + (\underline{\eta} \hat{\otimes} \underline{\eta})_{AB} - \frac{1}{2} \text{tr } \underline{\chi} \hat{\underline{\chi}}_{AB}, \end{aligned} \quad (2.4)$$

$$\begin{aligned} \nabla_3 \text{tr } \chi &= -\frac{1}{2} \text{tr } \chi \text{tr } \underline{\chi} + \not{g}^{AB} R_{3AB4} + 2\underline{\omega} \text{tr } \chi + 2\text{div } \eta + 2|\eta|^2 - \hat{\chi} \cdot \hat{\underline{\chi}} \\ &= -\text{tr } \chi \text{tr } \underline{\chi} + 2\underline{\omega} \text{tr } \chi + 2\text{div } \eta + 2|\eta|^2 - 2K + R + \text{Ric}_{34}, \\ \nabla_4 \text{tr } \underline{\chi} &= -\frac{1}{2} \text{tr } \chi \text{tr } \underline{\chi} + \not{g}^{AB} R_{3AB4} + 2\omega \text{tr } \underline{\chi} + 2\text{div } \underline{\eta} + 2|\underline{\eta}|^2 - \hat{\chi} \cdot \hat{\underline{\chi}} \\ &= -\text{tr } \chi \text{tr } \underline{\chi} + 2\omega \text{tr } \underline{\chi} + 2\text{div } \underline{\eta} + 2|\underline{\eta}|^2 - 2K + R + \text{Ric}_{34}. \end{aligned} \quad (2.5)$$

$$\begin{aligned} \nabla_4 \eta &= -\chi \cdot (\eta - \underline{\eta}) - \frac{1}{2} R_{A434}, \\ \nabla_3 \underline{\eta} &= -\underline{\chi} \cdot (\underline{\eta} - \eta) - \frac{1}{2} R_{A343}, \\ \nabla_4 \underline{\omega} &= \frac{1}{4} (\text{Ric}_{34} + \not{g}^{AB} R_{3AB4}) + 2\underline{\omega} \omega + \frac{1}{2} |\eta|^2 - \eta \cdot \underline{\eta} \\ &= 2\omega \underline{\omega} - \frac{1}{2} K + \frac{1}{2} \text{Ric}_{34} + \frac{1}{4} R + \frac{1}{4} \hat{\chi} \cdot \hat{\underline{\chi}} - \frac{1}{8} \text{tr } \chi \text{tr } \underline{\chi} + \frac{1}{2} |\eta|^2 - \eta \cdot \underline{\eta}, \\ \nabla_3 \omega &= \frac{1}{4} (\text{Ric}_{34} + \not{g}^{AB} R_{3AB4}) + 2\omega \omega + \frac{1}{2} |\underline{\eta}|^2 - \eta \cdot \underline{\eta} \\ &= 2\omega \omega - \frac{1}{2} K + \frac{1}{2} \text{Ric}_{34} + \frac{1}{4} R + \frac{1}{4} \hat{\chi} \cdot \hat{\underline{\chi}} - \frac{1}{8} \text{tr } \chi \text{tr } \underline{\chi} + \frac{1}{2} |\underline{\eta}|^2 - \eta \cdot \underline{\eta}. \end{aligned} \quad (2.6)$$

Moreover, we have equations for ζ :

$$\begin{aligned}
\Omega \text{Ric}_{3A} &= \Omega \nabla_3 \zeta + \frac{3}{2} \Omega \text{tr} \underline{\chi} \zeta + \Omega \underline{\hat{\chi}} \cdot \zeta + \Omega \text{tr} \underline{\chi} \nabla \log \Omega \\
&\quad + 2 \nabla(\Omega \underline{\omega}) + \text{div}(\Omega \underline{\hat{\chi}}) - \frac{1}{2} \nabla(\Omega \text{tr} \underline{\chi}), \\
\Omega \text{Ric}_{4A} &= -\Omega \nabla_4 \zeta - \frac{3}{2} \Omega \text{tr} \chi \zeta - \Omega \hat{\chi} \cdot \zeta + \Omega \text{tr} \chi \nabla \log \Omega \\
&\quad + 2 \nabla(\Omega \omega) + \text{div}(\Omega \hat{\chi}) - \frac{1}{2} \nabla(\Omega \text{tr} \chi).
\end{aligned} \tag{2.7}$$

Here K stands for Gaussian curvature of topological sphere $S_{u,v}$. Gauss-Codazzi equations imply

$$K = \frac{1}{2} \not\partial^{AB} R_{3A4B} + \frac{1}{2} R + \frac{1}{2} \text{Ric}_{34} + \frac{1}{2} \hat{\chi} \cdot \hat{\chi} - \frac{1}{4} \text{tr} \chi \text{tr} \underline{\chi}, \tag{2.8}$$

$$\begin{aligned}
\nabla^B \hat{\chi}_{AB} - \frac{1}{2} \nabla_A \text{tr} \chi &= -\frac{1}{2} R_{A434} + \text{Ric}_{4A} + \frac{1}{2} \text{tr} \chi \zeta_A - \zeta^B \hat{\chi}_{AB}, \\
\nabla^B \underline{\hat{\chi}}_{AB} - \frac{1}{2} \nabla_A \text{tr} \underline{\chi} &= -\frac{1}{2} R_{A343} + \text{Ric}_{3A} - \frac{1}{2} \text{tr} \underline{\chi} \zeta_A + \zeta^B \underline{\hat{\chi}}_{AB}.
\end{aligned} \tag{2.9}$$

$$\begin{aligned}
\nabla_A \eta_B - \nabla_B \eta_A &= R_{4[AB]3} + \hat{\chi}^C_{[A} \hat{\chi}_{B]C}, \\
\nabla_A \underline{\eta}_B - \nabla_B \underline{\eta}_A &= -R_{4[AB]3} - \hat{\chi}^C_{[A} \hat{\chi}_{B]C},
\end{aligned} \tag{2.10}$$

For Weyl curvature,

$$W_{\mu\nu\theta\lambda} = R_{\mu\nu\theta\lambda} - \frac{1}{2} (\text{Ric}_{\mu\theta} g_{\nu\lambda} + g_{\mu\theta} \text{Ric}_{\nu\lambda} - \text{Ric}_{\nu\theta} g_{\mu\lambda} - g_{\nu\theta} \text{Ric}_{\mu\lambda}) + \frac{R}{6} (g_{\mu\theta} g_{\nu\lambda} - g_{\nu\theta} g_{\mu\lambda}),$$

we define the curvature components

$$\begin{aligned}
\beta_A &= \frac{1}{2} W(e_A, e_4, e_3, e_4), \quad \underline{\beta}_A = \frac{1}{2} W(e_A, e_3, e_3, e_4), \\
\rho &= \frac{1}{4} W(e_4, e_3, e_4, e_3), \quad \sigma = \frac{1}{4} {}^* W(e_4, e_3, e_4, e_3).
\end{aligned} \tag{2.11}$$

Then previous equations imply the following algebraic relations,

$$K = -\rho + \frac{1}{2} R + \text{Ric}_{34} + \frac{1}{2} \hat{\chi} \cdot \hat{\chi} - \frac{1}{4} \text{tr} \underline{\chi} \text{tr} \chi, \tag{2.12}$$

$$\text{curl} \eta = -\text{curl} \underline{\eta} = \sigma + \frac{1}{2} \hat{\chi} \wedge \hat{\chi}, \tag{2.13}$$

$$\text{div} \hat{\chi} - \frac{1}{2} \nabla \text{tr} \chi = -\beta + \frac{1}{2} \text{Ric}_4 + \frac{1}{2} \text{tr} \chi \zeta - \zeta \cdot \hat{\chi}, \tag{2.14}$$

$$\text{div} \underline{\hat{\chi}} - \frac{1}{2} \nabla \text{tr} \underline{\chi} = \underline{\beta} + \frac{1}{2} \text{Ric}_3 - \frac{1}{2} \text{tr} \underline{\chi} \zeta + \zeta \cdot \underline{\hat{\chi}}. \tag{2.15}$$

After introducing the reduced curvature

$$\beta_A^r = \beta_A - \frac{1}{2} \text{Ric}_{4A}, \quad \underline{\beta}_A^r = \underline{\beta}_A + \frac{1}{2} \text{Ric}_{3A}, \quad \sigma^r = \sigma + \frac{1}{2} \hat{\chi} \wedge \hat{\chi}, \tag{2.16}$$

we can write the equations for curvature components

$$\begin{aligned}
\Omega \nabla_3 K + \Omega \text{tr} \underline{\chi} K &= \text{div}(\Omega \underline{\beta}^r) + \frac{1}{2} \text{div}(\eta \Omega \text{tr} \underline{\chi}) - \text{div}(\eta \cdot \Omega \underline{\hat{\chi}}), \\
\Omega \nabla_4 K + \Omega \text{tr} \chi K &= -\text{div}(\Omega \beta^r) + \frac{1}{2} \text{div}(\underline{\eta} \Omega \text{tr} \chi) - \text{div}(\underline{\eta} \cdot \Omega \hat{\chi}),
\end{aligned} \tag{2.17}$$

$$\begin{aligned}
\nabla_3 \beta^r + (\text{tr } \underline{\chi} - 2\underline{\omega}) \beta^r &= -\nabla K + {}^* \nabla \sigma + 2\widehat{\chi} \cdot \underline{\beta} \\
&\quad + \frac{1}{2} (\nabla(\widehat{\chi} \cdot \widehat{\chi}) - {}^* \nabla(\widehat{\chi} \wedge \widehat{\chi})) - \frac{1}{4} \nabla (\text{tr } \chi \text{tr } \underline{\chi}) \\
&\quad + 3(\eta\rho + {}^* \eta\sigma) + \nabla_A (g^{CD} \text{Ric}_{CD}) - \nabla^B \text{Ric}_{BA} - \frac{1}{2} \text{tr } \underline{\chi} \text{Ric}_{4A}, \\
\nabla_4 \underline{\beta}^r + (\text{tr } \chi - 2\omega) \underline{\beta}^r &= \nabla K + {}^* \nabla \sigma + 2\widehat{\chi} \cdot \beta \\
&\quad - \frac{1}{2} (\nabla(\widehat{\chi} \cdot \widehat{\chi}) + {}^* \nabla(\widehat{\chi} \wedge \widehat{\chi})) + \frac{1}{4} \nabla (\text{tr } \chi \text{tr } \underline{\chi}) \\
&\quad - 3(\underline{\eta}\rho - {}^* \underline{\eta}\sigma) - \nabla_A (g^{CD} \text{Ric}_{CD}) + \nabla^B \text{Ric}_{BA} + \frac{1}{2} \text{tr } \chi \text{Ric}_{3A}.
\end{aligned} \tag{2.18}$$

Using (2.6), we derive

$$\begin{aligned}
\Omega \nabla_3 \text{curl } \underline{\eta} + \Omega \text{tr } \underline{\chi} \text{curl } \underline{\eta} &= \text{curl}(\Omega \underline{\beta}^r) + \epsilon^{AC} \nabla_A (\Omega \underline{\chi} \cdot \eta)_C - \epsilon^{AC} \nabla_A (\Omega \text{Ric})_{C3}, \\
\Omega \nabla_4 \text{curl } \eta + \Omega \text{tr } \chi \text{curl } \eta &= -\text{curl}(\Omega \beta^r) + \epsilon^{AC} \nabla_A (\Omega \chi \cdot \underline{\eta})_C - \epsilon^{AC} \nabla_A (\Omega \text{Ric})_{C4},
\end{aligned} \tag{2.19}$$

and

$$\begin{aligned}
\Omega \nabla_3 (\text{div } \underline{\eta} - K) + \Omega \text{tr } \underline{\chi} (\text{div } \underline{\eta} - K) &= 4 \text{div}(\Omega \widehat{\chi} \cdot \zeta) - \text{div}(\Omega \text{Ric}_3), \\
\Omega \nabla_4 (\text{div } \eta - K) + \Omega \text{tr } \chi (\text{div } \eta - K) &= -4 \text{div}(\Omega \widehat{\chi} \cdot \zeta) - \text{div}(\Omega \text{Ric}_4).
\end{aligned} \tag{2.20}$$

For a scalar function ϕ , there is equation

$$-\Omega^2 \square \phi = \Omega e_3(\Omega e_4 \phi) + \frac{1}{2} \Omega \text{tr } \chi \Omega e_3 \phi + \frac{1}{2} \Omega \text{tr } \underline{\chi} \Omega e_4 \phi - \Omega^2 \Delta \phi - 2\Omega^2 \eta \cdot \nabla \phi. \tag{2.21}$$

2.2 Local existence

We now introduce some results on Einstein vacuum equations. Similar results can be derived for other Einstein field equations and we omit them due to length of the article.

We first state the existence theorem for characteristic initial value problem of Einstein vacuum equations from [3].

Theorem 2.1. *Given initial null hypersurface $H_0, \underline{H}_0$ and regular initial metric (g, Ω, b) on it, we define $\chi_{AB} = \frac{1}{2} \mathcal{L}_{e_4} g_{AB}$ on H_0 and $\underline{\chi}_{AB} = \frac{1}{2} \mathcal{L}_{e_3} g_{AB}$ on $\underline{H}_0$. Suppose*

$$e_4(\text{tr } \chi) + \frac{1}{2}(\text{tr } \chi)^2 + |\widehat{\chi}|^2 + 2\omega \text{tr } \chi = 0, \quad e_3(\text{tr } \underline{\chi}) + \frac{1}{2}(\text{tr } \underline{\chi})^2 + |\widehat{\underline{\chi}}|^2 + 2\underline{\omega} \text{tr } \underline{\chi},$$

hold on $H_0, \underline{H}_0$ respectively, then there is a future open neighborhood of $H_0 \cup \underline{H}_0$ and metric (g, b, Ω) on it solving the Einstein vacuum equations and satisfying the initial data.

This theorem reveals the local existence, and we further remark that for larger region, such as $(0, 1)^2 \times \mathbb{S}^2$, as long as the a priori estimates hold in it, the solution can be extended to the region.

2.3 Residual control

In this paper, we target to deal with asymmetric Einstein equations, which in general do not have explicit solutions. To measure the computational error, we use Ricci residues. For Einstein equation

$$G_{\mu\nu} := \text{Ric}_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = T_{\mu\nu},$$

we measure $\|G - T\|_{L^2(S_{u,v})}$ to evaluate the approximation. The theorems below reveals that in our case the difference of metrics is controlled by the difference of curvature residues.

Theorem 2.2. We consider metric $\overset{\circ}{g}_{AB}, \overset{\circ}{b}, \overset{\circ}{\Omega}$ on $(0, 1) \times (0, 1) \times \mathbb{S}^2$ such that

$$\overset{\circ}{\text{Ric}}_{44}(0, v) = 0, \quad \overset{\circ}{\text{Ric}}_{33}(u, 0) = 0, \quad \overset{\circ}{\text{Ric}}_{4A}(0, v) = 0, \quad \overset{\circ}{\text{Ric}}_{3A}(u, 0) = 0,$$

and estimates

$$\|\overset{\circ}{\text{Ric}}_{AB}, \overset{\circ}{\text{Ric}}_3, \overset{\circ}{\text{Ric}}_4, \overset{\circ}{\text{Ric}}_{34}, \overset{\circ}{\text{Ric}}_{33}, \overset{\circ}{\text{Ric}}_{44}\|_{L_u^\infty L_v^\infty H^3(S)} \leq \epsilon. \quad (2.22)$$

We assume moreover

$$\|\partial_{\mathbb{S}} \overset{\circ}{g}, \partial_{\mathbb{S}} \overset{\circ}{b}, \partial_{\mathbb{S}} \overset{\circ}{\Omega}, \psi\|_{L_u^\infty L_v^\infty H^4(S)} \leq C_0, \quad (2.23)$$

where ψ stands for Ricci coefficients in interest. Then for $\epsilon = \epsilon(C_0)$ sufficiently small, there exists (g, b, Ω) solving EVE on $(0, 1) \times (0, 1) \times \mathbb{S}^2$ with initial data

$$(g, b, \Omega) = (\overset{\circ}{g}, \overset{\circ}{b}, \overset{\circ}{\Omega}), \quad \text{on } \underline{H}_0 \cup H_0,$$

and estimates

$$\|g - \overset{\circ}{g}, b - \overset{\circ}{b}, \Omega - \overset{\circ}{\Omega}\|_{L_u^\infty L_v^\infty H^4(S)} \leq \epsilon^{1/2}. \quad (2.24)$$

The methods of deriving the equations of difference and doing estimates are similar to section 7 of [2], so we only write the proof in sketch.

Define auxiliary functions $\omega^\dagger, \underline{\omega}^\dagger$ by

$$\nabla_3 \omega^\dagger = \frac{1}{2} \text{curl } \zeta, \quad \omega^\dagger|_{H_{-1}} = 0, \quad \nabla_4 \underline{\omega}^\dagger = \frac{1}{2} \text{curl } \zeta, \quad \underline{\omega}^\dagger|_{\underline{H}_0} = 0.$$

We make bootstrap assumption

$$\begin{aligned} & \|\widetilde{\beta}, \widetilde{K}, \nabla \widetilde{\eta}, \nabla \widetilde{\eta}, \nabla \widetilde{\chi}, \nabla \widetilde{\omega}, \nabla \widetilde{\omega}^\dagger\|_{L_v^2 H^3(S)} \\ & + \|\widetilde{K}, \widetilde{\beta}, \nabla \widetilde{\eta}, \nabla \widetilde{\eta}, \nabla \widetilde{\chi}, \nabla \widetilde{\omega}, \nabla \widetilde{\omega}^\dagger\|_{L_u^2 H^3(S)} \leq B \exp(D(v+u))\epsilon. \end{aligned}$$

We will prove that we can choose D sufficiently large so that the bootstrap constant B can be improved to $B \leq 1$.

The motivation is direct. Let $\partial_u f \leq C_0 f, \partial_v f \leq C_0 f$, and suppose

$$f(u, v) \leq B \exp(D(v+u))\epsilon,$$

then we integrate it

$$\begin{aligned} f(u, v) - f(0, v) & \leq C_0 \int_0^u f(u', v) du' \\ & \leq \frac{C_0 B}{D} (\exp(D(u+v)) - \exp(Dv)) \epsilon \leq \frac{C_0 B}{D} \exp(D(u+v)) \epsilon, \\ f(u, v) - f(u, 0) & \leq C_0 \int_0^v f(u, v') dv' \\ & \leq \frac{C_0 B}{D} (\exp(D(u+v)) - \exp(Du)) \epsilon \leq \frac{C_0 B}{D} \exp(D(u+v)) \epsilon. \end{aligned}$$

If we choose $D^{1/2} \geq C_0 B$, the bootstrap assumption will be closed. Absorbing the bootstrap constant by the parameter D actually shows the increasing rate is controlled exponentially. Even though the norms increase quickly, they will not exceed $O(\epsilon) \leq o(\epsilon^{1/2})$ in finite region.

In summary, we conclude

Lemma 2.3. Suppose $\|f\|_{L^2(S_{u,v})} \leq B \exp(D(v+u))\epsilon$, then we have

$$\|f\|_{L_u^2 L^2(S)} + \|f\|_{L_v^2 L^2(S)} \leq D^{-1/2} \exp(D(v+u))\epsilon, \quad (2.25)$$

when D is sufficiently large.

Suppose either $\|h\|_{L_u^2 L^2(S)} \leq B \exp(D(u+v))\epsilon$ or $\|h\|_{L_v^2 L^2(S)} \leq B \exp(D(u+v))\epsilon$, then we have

$$\|h\|_{L_u^2 L_v^2 L^2(S)} \leq D^{-1/2} \exp(D(v+u))\epsilon, \quad (2.26)$$

when D is sufficiently large.

We use notation

$$\psi_1 \in \{\eta, \underline{\eta}, \chi, \omega\}, \quad \psi_2 \in \{\eta, \underline{\eta}, \underline{\chi}, \underline{\omega}\}$$

Because

$$\mathcal{L}_v b = -2\Omega^2(\eta - \underline{\eta}), \quad \mathcal{L}_v g = 2\Omega\chi, \quad \mathcal{L}_v \log \Omega = -2\Omega\omega,$$

we obtain that

$$\|\widetilde{b}, \widetilde{g}, \widetilde{\log \Omega}\|_{H^4(S_{u,v})} \lesssim \|\widetilde{\eta}, \widetilde{\underline{\eta}}, \widetilde{\chi}, \widetilde{\omega}, \widetilde{\Omega}, \widetilde{g}\|_{L_v^2 H^4(S)} \lesssim B \exp(D(u+v))\epsilon. \quad (2.27)$$

For Bianchi pair (Ψ_1, Ψ_2) in interest, we consider equations

$$\begin{aligned} \Omega \nabla_3 \widetilde{\Psi}_1 + \mathcal{D} \widetilde{\Psi}_2 &= \widetilde{\psi} \widetilde{\Psi} + \widetilde{\psi} \nabla \widetilde{\psi} + \widetilde{\psi}^3 + \widetilde{b} \cdot \nabla \overset{\circ}{\Psi}_1 + \widetilde{\partial g} \overset{\circ}{\Psi}_2 + \widetilde{g} \nabla \overset{\circ}{\Psi}_2 + \overset{\circ}{\psi} \overset{\circ}{\text{Ric}} + \nabla \overset{\circ}{\text{Ric}}, \\ \Omega \nabla_4 \widetilde{\Psi}_2 - {}^* \mathcal{D} \widetilde{\Psi}_1 &= \widetilde{\psi} \widetilde{\Psi} + \widetilde{\psi} \nabla \widetilde{\psi} + \widetilde{\psi}^3 + \widetilde{\partial g} \overset{\circ}{\Psi}_1 + \widetilde{g} \nabla \overset{\circ}{\Psi}_1 + \overset{\circ}{\psi} \overset{\circ}{\text{Ric}} + \nabla \overset{\circ}{\text{Ric}}. \end{aligned}$$

Energy estimates give that

$$\begin{aligned} \|\widetilde{\Psi}_1\|_{L_v^2 H^3(S)}^2 + \|\widetilde{\Psi}_2\|_{L_u^2 H^3(S)}^2 \\ \lesssim \epsilon^2 + \|\widetilde{\partial g}, \widetilde{g}, \widetilde{\psi}, \widetilde{\nabla \psi}, \widetilde{\Psi}, \widetilde{b}\|_{L_u^2 L_v^2 H^3(S)}^2 \lesssim (1 + D^{-1/2}) \exp(2D(u+v))\epsilon^2. \end{aligned}$$

Applying the estimate to pairs

$$(\beta, (K, \text{curl } \eta)), \quad ((K, \text{curl } \underline{\eta}), \underline{\beta}), \quad (\nabla \text{tr } \underline{\chi}, 0), \quad (0, \nabla \text{tr } \chi),$$

$$(\text{div } \underline{\eta} - K, 0), \quad (0, \text{div } \eta - K), \quad (\nabla \omega + {}^* \nabla \omega^\dagger - \frac{1}{2} \beta, 0), \quad (0, -\nabla \underline{\omega} + {}^* \nabla \underline{\omega}^\dagger - \frac{1}{2} \underline{\beta}),$$

will finish the proof.

Remark 2.4. The conditions

$$\overset{\circ}{\text{Ric}}_{44}(-1, v) = 0, \quad \overset{\circ}{\text{Ric}}_{33}(u, 0) = 0,$$

guarantee that $(\overset{\circ}{g}, \overset{\circ}{b}, \overset{\circ}{\Omega})$ is a valid initial data set for characteristic initial value problem of Einstein vacuum equation, and

$$\overset{\circ}{\text{Ric}}_{4A}(-1, v) = 0, \quad \overset{\circ}{\text{Ric}}_{3A}(u, 0) = 0,$$

ensure that $\overset{\circ}{\zeta}$ along H_{-1} and $\underline{H}_0$ is equal to the ζ of corresponding EVE solution.

3 Picard Iteration Design and Numerical Strategies

In this section, we introduce the iteration design and numerical strategies.

3.1 Picard iteration for Einstein equations

In this section, we first consider Einstein scalar-field equations

$$\text{Ric} = d\phi \otimes d\phi, \quad \square_g \phi = 0.$$

We work in region

$$Q = \{-1 < u < u_0, \quad 0 < v < v_0(u)\}.$$

The first task is to construct a valid initial data for ESE. We directly prescribe g, b, Ω along $\underline{H}_0$. The only requirement is

$$-\Omega \nabla_3(\Omega \text{tr} \underline{\chi}) - \frac{1}{2}(\Omega \text{tr} \underline{\chi})^2 - |\Omega \widehat{\underline{\chi}}|^2 - 4\Omega \underline{\omega} \Omega \text{tr} \underline{\chi} \geq 0$$

Then $\Omega e_3 \phi(u, 0)$ can be assigned via equation

$$\Omega e_3 \phi = \left(-\Omega \nabla_3(\Omega \text{tr} \underline{\chi}) - \frac{1}{2}(\Omega \text{tr} \underline{\chi})^2 - |\Omega \widehat{\underline{\chi}}|^2 - 4\Omega \underline{\omega} \Omega \text{tr} \underline{\chi} \right)^{1/2}.$$

After defining $\phi(-1, 0)$ and $\zeta(-1, 0)$, we derive $\phi(u, 0)$ from $\Omega e_3 \phi(u, 0)$ and $\zeta(u, 0)$ via

$$\Omega \nabla_3 \zeta + \frac{3}{2} \Omega \text{tr} \underline{\chi} \zeta + \Omega \widehat{\underline{\chi}} \cdot \zeta = -2\nabla(\Omega \underline{\omega}) - \text{div}(\Omega \widehat{\underline{\chi}}) + \frac{1}{2} \nabla(\Omega \text{tr} \underline{\chi}) - \Omega \text{tr} \underline{\chi} \nabla \log \Omega + \Omega e_3 \phi \nabla \phi.$$

Thus $\eta = \zeta + \nabla \log \Omega$ and $\eta = -\zeta + \nabla \log \Omega$ are also prescribed. We then define $\Omega \text{tr} \chi(-1, 0)$ and $\Omega \widehat{\chi}(-1, 0)$, and compute $\Omega \chi(u, 0)$ by

$$\Omega \nabla_3(\Omega \text{tr} \chi) + \Omega \text{tr} \underline{\chi} \Omega \text{tr} \chi = \Omega^2 \left(2 \text{div} \eta + 2 |\eta|^2 - 2K(g) + |\nabla \phi|^2 \right),$$

$$\Omega \nabla_3(\Omega \widehat{\chi}) + \frac{1}{2} \Omega \text{tr} \underline{\chi} \Omega \widehat{\chi} = \Omega^2 (\nabla \hat{\otimes} \eta + \eta \hat{\otimes} \eta) - \frac{1}{2} \Omega \text{tr} \chi \Omega \widehat{\underline{\chi}} + \frac{1}{2} \Omega^2 \nabla \phi \hat{\otimes} \nabla \phi.$$

We also directly assign $\Omega e_4 \phi(-1, 0)$ and $\Omega \omega(-1, 0)$, and compute them along $\underline{H}_0$ via

$$\Omega e_3(\Omega e_4 \phi) + \frac{1}{2} \Omega \text{tr} \chi \Omega e_3 \phi + \frac{1}{2} \Omega \text{tr} \underline{\chi} \Omega e_4 \phi - \Omega^2 \Delta \phi - 2\Omega^2 \eta \cdot \nabla \phi = 0,$$

$$\frac{1}{2} \Omega e_3 \phi \Omega e_4 \phi + \frac{1}{2} \Omega^2 |\nabla \phi|^2 = 2\Omega \nabla_3(\Omega \omega) + \Omega^2 K - \frac{1}{2} \Omega \widehat{\underline{\chi}} \cdot \Omega \widehat{\underline{\chi}} + \frac{1}{4} \Omega \text{tr} \chi \Omega \text{tr} \underline{\chi} - \Omega^2 |\underline{\eta}|^2 + 2\Omega^2 \eta \cdot \underline{\eta}.$$

Along H_{-1} , we prescribe $\Omega e_4 \phi(-1, v)$ and $\Omega \omega(-1, v)$ to compute ϕ, Ω . Consider a symmetric and $g(-1, 0)$ -trace-free tensor $X_{AB}(-1, v)$. We then prescribe $g, \Omega \chi$ along H_{-1} by solving:

$$\begin{cases} \Omega \widehat{\chi}_{AB} = \frac{1}{2} \left(X_{AC} (g(-1, 0))^{CD} g_{DB} + X_{BC} (g(-1, 0))^{CD} g_{DA} \right), \\ \partial_v(\Omega \text{tr} \chi) = -\frac{1}{2}(\Omega \text{tr} \chi)^2 - 4\Omega \omega \Omega \text{tr} \chi - (\Omega e_4 \phi)^2 - \Omega \widehat{\chi}_{AB} \Omega \widehat{\chi}_{CD} g^{AC} g^{BD}, \\ \mathcal{L}_v g_{AB} = \Omega \text{tr} \chi g_{AB} + 2\Omega \widehat{\chi}_{AB}. \end{cases} \quad (3.1)$$

With these initial data, we can start the iteration. The quantities to be iterated are

$$g^{(i)}, (b^{(i)})^A, \log \Omega^{(i)}, (\Omega \chi)_{AB}^{(i)}, (\Omega \underline{\chi})_{AB}^{(i)}, (\zeta^{(i)})^A, (\Omega \omega)^{(i)}, (\Omega \underline{\omega})^{(i)}, (\Omega e_3 \phi)^{(i)}, (\Omega e_4 \phi)^{(i)}, \nabla_A \phi^{(i)}.$$

Initially, we set $g^{(0)} = g|_{v=0}$, $\Omega^{(0)} = \Omega|_{v=0}$, $\zeta^{(0)} = \zeta|_{v=0}$,

$$(b^{(0)})^A(u, v) = b^A(u, 0) - \int_0^v 4(\Omega^{(0)})^2 (\zeta^{(0)})^A(u, v') \, dv',$$

$\widehat{\chi}^{(0)} = \widehat{\chi}|_{v=0}$, $\text{tr } \chi^{(0)} = \text{tr } \chi|_{v=0}$, $\widehat{\underline{\chi}}^{(0)} = \widehat{\underline{\chi}}|_{v=0}$, $\text{tr } \underline{\chi}^{(0)} = \text{tr } \underline{\chi}|_{v=0}$, $\omega^{(0)} = \omega|_{v=0}$, $\underline{\omega}^{(0)} = \underline{\omega}|_{v=0}$, $(e_3\phi)^{(0)} = e_3\phi|_{v=0}$, $(e_4\phi)^{(0)} = e_4\phi|_{v=0}$, $(\nabla_A\phi)^{(0)} = \nabla_A\phi|_{v=0}$. For all i , we always require

$$\eta_A^{(i)} = (\zeta^{(i)})^B g_{AB}^{(i)} + \nabla_A \log \Omega^{(i)}, \quad \underline{\eta}_A^{(i)} = -(\zeta^{(i)})^B g_{AB}^{(i)} + \nabla_A \log \Omega^{(i)}$$

1. Construction of $(\Omega\widehat{\chi})^{(i+1/2)}$, $\partial_v\phi^{(i+1)}$, and $\nabla_A\phi^{(i+1)}$

We use (2.4) to define $(\Omega\widehat{\chi})^{(i+1/2)}$,

$$\left\{ \begin{aligned} & \mathcal{L}_{\partial_u+b^{(i)}}(\Omega\widehat{\chi}^{(i+1/2)})_{AB} - \frac{1}{2}(\Omega \text{tr } \underline{\chi})^{(i)}(\Omega\widehat{\chi}^{(i+1/2)})_{AB} \\ & \quad - (\Omega\widehat{\underline{\chi}}^{(i)})_A{}^C (\Omega\widehat{\chi}^{(i+1/2)})_{CB} - (\Omega\widehat{\underline{\chi}}^{(i)})_B{}^C (\Omega\widehat{\chi}^{(i+1/2)})_{CA} \\ & = (\Omega^{(i)})^2 \left(\left(\nabla^{(i)} \hat{\otimes} \eta^{(i)} + \eta^{(i)} \hat{\otimes} \nabla^{(i)} + \frac{1}{2} \nabla\phi^{(i)} \hat{\otimes} \nabla\phi^{(i)} \right)_{AB} - \frac{1}{2}(\text{tr } \chi)^{(i)} \widehat{\underline{\chi}}_{AB}^{(i)} \right), \\ & \Omega\widehat{\chi}^{(i+1/2)}_{AB}(-1, v) = \frac{1}{2} \left(\Omega\widehat{\chi}_{AC} g^{CD} g^{(i)}_{DB} + \Omega\widehat{\chi}_{BC} g^{CD} g^{(i)}_{DA} \right) (-1, v). \end{aligned} \right. \quad (3.2)$$

Using (2.21), we define $\partial_v\phi^{(i+1)}$ by

$$\left\{ \begin{aligned} & (\Omega e_3)^{(i)}(\partial_v\phi^{(i+1)}) + \frac{1}{2}(\Omega \text{tr } \underline{\chi})^{(i)}(\partial_v\phi^{(i+1)}) \\ & = (\Omega^{(i)})^2 \Delta^{(i)}\phi^{(i)} - \frac{1}{2}(\Omega \text{tr } \chi)^{(i)}(\Omega e_3)^{(i)}\phi^{(i)} + 2(\Omega^{(i)})^2 \eta^{(i)} \nabla\phi^{(i)}, \\ & \partial_v\phi^{(i+1)}(-1, v) = \partial_v\phi(-1, v). \end{aligned} \right. \quad (3.3)$$

It follows that

$$\nabla_A\phi^{(i+1)}(u, v) = \nabla_A\phi(u, 0) + \partial_A \left(\int_0^v \partial_v\phi^{(i+1)} \right).$$

2. Construction of $(\Omega\underline{\omega})^{(i+1/2)}$

We use (2.6) to define $(\Omega\underline{\omega})^{(i+1/2)}$,

$$\left\{ \begin{aligned} & 4\partial_v(\Omega\underline{\omega})^{(i+1/2)} = \Omega\widehat{\chi}_{AB}^{(i+1/2)} \Omega\widehat{\underline{\chi}}_{CD}^{(i)} (g^{(i)})^{AC} (g^{(i)})^{BD} + \frac{1}{2}(\Omega \text{tr } \chi)^{(i)} (\Omega \text{tr } \underline{\chi})^{(i)} \\ & \quad - 4(\Omega^{(i)})^2 \eta^{(i)} \cdot \underline{\eta}^{(i)} + (\partial_u + b^{(i)})(\Omega \text{tr } \chi)^{(i)} - 2(\Omega^{(i)})^2 \text{div}^{(i)} \eta^{(i)} \\ & \quad + (\partial_v\phi)^{(i+1)}(\Omega e_3\phi)^{(i)}, \\ & (\Omega\underline{\omega})^{(i+1/2)}(u, 0) = \Omega\underline{\omega}(u, 0). \end{aligned} \right. \quad (3.4)$$

3. Construction of $\Omega^{(i+1)}$, $\nabla_A \log \Omega^{(i+1)}$, and $(\Omega\omega)^{(i+1)}$

With $(\Omega\underline{\omega})^{(i+1/2)}$, we can integrate it to obtain $\Omega^{(i+1)}$

$$\left\{ \begin{aligned} & \left(\partial_u + b^{(i)} \right) \log \Omega^{(i+1)} = -2(\Omega\underline{\omega})^{(i+1/2)}, \\ & \log \Omega^{(i+1)}(-1, v) = \log \Omega(-1, v), \end{aligned} \right. \quad (3.5)$$

$$\nabla_A \log \Omega^{(i+1)} = \partial_A \log \Omega^{(i+1)}. \quad (3.6)$$

We construct $(\Omega\omega)^{(i+1)}$ by integration instead of differentiation,

$$\left\{ \begin{aligned} & 4 \left(\partial_u + b^{(i)} \right) (\Omega\omega)^{(i+1)} = \Omega\widehat{\chi}_{AB}^{(i+1/2)} \Omega\widehat{\underline{\chi}}_{CD}^{(i)} (g^{(i)})^{AC} (g^{(i)})^{BD} + \frac{1}{2}(\Omega \text{tr } \chi)^{(i)} (\Omega \text{tr } \underline{\chi})^{(i)} \\ & \quad - 4(\Omega^{(i)})^2 \eta^{(i)} \cdot \underline{\eta}^{(i)} + (\partial_u + b^{(i)})(\Omega \text{tr } \chi)^{(i)} - 2(\Omega^{(i)})^2 \text{div}^{(i)} \eta^{(i)} \\ & \quad + (\partial_v\phi)^{(i+1)}(\Omega e_3\phi)^{(i)} - 8(\Omega^{(i)})^2 (\zeta^{(i)})^A \nabla_A \log \Omega^{(i+1)}, \\ & (\Omega\omega)^{(i+1)}(-1, v) = \Omega\omega(-1, v). \end{aligned} \right. \quad (3.7)$$

We note that the construction of $(\Omega\omega)^{(i+1)}$ is equivalent to $(\Omega\omega)^{(i+1)} = -\frac{1}{2}\partial_v \log \Omega^{(i+1)}$, because

$$\begin{aligned} 4(\Omega e_3)^{(i)}(\Omega\omega)^{(i+1)} &= -2\partial_v(\Omega e_3)^{(i)} \log \Omega^{(i+1)} + 2\mathcal{L}_v b^{(i)} \nabla \log \Omega^{(i+1)} \\ &= 4\partial_v(\Omega\omega)^{(i+1/2)} - 8(\Omega^{(i)})^2(\zeta^{(i)})^A \nabla_A \log \Omega^{(i+1)}. \end{aligned} \quad (3.8)$$

4. Construction of $\zeta^{(i+1)}$, $b^{(i+1)}$, and $(\Omega\omega)^{(i+1)}$

From (2.7), we define $\zeta^{(i+1)}$,

$$\left\{ \begin{aligned} \mathcal{L}_v(\zeta^{(i+1)})^A &= -2(\Omega \operatorname{tr} \chi)^{(i)}(\zeta^{(i)})^A - 2(g^{(i)})^{AC} \left((\Omega\widehat{\chi})^{(i+1/2)} \right)_{CB} (\zeta^{(i)})^B \\ &\quad + 2(g^{(i)})^{AB} \partial_B(\Omega\omega)^{(i+1)} + (\operatorname{div})^{(i)}(\Omega\widehat{\chi})_B^{(i+1/2)} (g^{(i)})^{BA} \\ &\quad - \frac{1}{2}(g^{(i)})^{AB} \partial_B(\Omega \operatorname{tr} \chi)^{(i)} + (\Omega \operatorname{tr} \chi)^{(i)} (g^{(i)})^{AB} \partial_B \log \Omega^{(i+1)} \\ &\quad - (\partial_v \phi)^{(i+1)} \partial_B \phi^{(i)} (g^{(i)})^{AB}, \\ (\zeta^{(i+1)})^A(u, 0) &= \zeta^A(u, 0). \end{aligned} \right. \quad (3.9)$$

We remark that it is necessary to treat $\zeta^{(i+1)}$ as a vector field instead of a 1-form in order to define $b^{(i+1)}$,

$$\left\{ \begin{aligned} \mathcal{L}_v(b^{(i+1)})^A &= -4(\Omega^{(i+1)})^2(\zeta^{(i+1)})^A, \\ (b^{(i+1)})^A(u, 0) &= b^A(u, 0). \end{aligned} \right. \quad (3.10)$$

5. Construction of $g^{(i+1)}$ and $(\Omega\chi)^{(i+1)}$

It remains to define $g^{(i+1)}$. We consider the following $(g, \operatorname{tr} \chi, \widehat{\chi})$ system,

$$\left\{ \begin{aligned} \mathcal{L}_v(g^{(i+1)})_{AB} &= (\Omega \operatorname{tr} \chi)^{(i+1)}(g^{(i+1)})_{AB} + 2(\Omega\widehat{\chi})^{(i+1)}_{AB}, \\ \partial_v(\Omega \operatorname{tr} \chi)^{(i+1)} &= -\frac{1}{2} \left((\Omega \operatorname{tr} \chi)^{(i+1)} \right)^2 - 4(\Omega\omega)^{(i+1)}(\Omega \operatorname{tr} \chi)^{(i+1)} \\ &\quad - (\Omega\widehat{\chi})^{(i+1)}_{AB}(\Omega\widehat{\chi})^{(i+1)}_{CD} g^{(i+1)AC} g^{(i+1)BD} - (\partial_v \phi)^{(i+1)2}, \\ (\Omega\widehat{\chi})^{(i+1)}_{AB} &= \frac{1}{2} \left((\Omega\widehat{\chi})^{(i+1/2)}_{AC} (g^{(i)})^{CD} g^{(i+1)}_{DB} + (\Omega\widehat{\chi})^{(i+1/2)}_{BC} (g^{(i)})^{CD} g^{(i+1)}_{DA} \right), \\ (g^{(i+1)})_{AB}(u, 0) &= g_{AB}(u, 0), \quad (\Omega \operatorname{tr} \chi)^{(i+1)}(u, 0) = \Omega \operatorname{tr} \chi(u, 0). \end{aligned} \right. \quad (3.11)$$

6. Construction of $(\Omega\chi)^{(i+1)}$ and $(\Omega e_3\phi)^{(i+1)}$

There are two equivalent ways to define $(\Omega\chi)^{(i+1)}$. The first one is using $\Omega\chi_{AB} = \frac{1}{2}\mathcal{L}_{\Omega e_3} g_{AB}$. We use the second way, which is using equation

$$\left\{ \begin{aligned} \mathcal{L}_v(\Omega\chi)^{(i+1)}_{AB} &= \mathcal{L}_{\partial_u + b^{(i+1)}}(\Omega\chi)^{(i+1)}_{AB} - 2(\Omega^{(i+1)})^2 \left(\nabla_A^{(i+1)} \zeta_B^{(i+1)} + \nabla_B^{(i+1)} \zeta_A^{(i+1)} \right) \\ &\quad - 4(\Omega^{(i+1)})^2 \left(\zeta_A^{(i+1)} \nabla_B \log \Omega^{(i+1)} + \zeta_B^{(i+1)} \nabla_A \log \Omega^{(i+1)} \right), \\ (\Omega\chi)^{(i+1)}_{AB}(u, 0) &= \Omega\chi_{AB}(u, 0). \end{aligned} \right. \quad (3.12)$$

We remark that

$$\begin{aligned} &\Omega \nabla_4(\Omega\chi)_{AB} - \Omega \nabla_3(\Omega\chi)_{AB} \\ &= \mathcal{L}_v(\Omega\chi)_{AB} - (\Omega\chi)_A^C (\Omega\chi)_{CB} - (\Omega\chi)_B^C (\Omega\chi)_{CA} \\ &\quad - \mathcal{L}_{\partial_u + b}(\Omega\chi)_{AB} + (\Omega\chi)_A^C (\Omega\chi)_{CB} + (\Omega\chi)_B^C (\Omega\chi)_{AC} \\ &= \mathcal{L}_v(\Omega\chi)_{AB} - \mathcal{L}_{\partial_u + b}(\Omega\chi)_{AB}. \end{aligned} \quad (3.13)$$

Theoretically, $(\Omega\chi)^{(i+1)}_{AB} = \frac{1}{2}\mathcal{L}_{\partial_u + b^{(i+1)}} g_{AB}^{(i+1)}$. Similarly, we construct $(\Omega e_3\phi)^{(i+1)}$ by

$$\left\{ \begin{aligned} \partial_v(\Omega e_3\phi)^{(i+1)} &= (\partial_u + b^{(i+1)})\partial_v \phi^{(i+1)} - 4(\Omega^{(i+1)})^2(\zeta^{(i+1)})^A \nabla_A \phi^{(i+1)}, \\ (\Omega e_3\phi)^{(i+1)}(u, 0) &= \Omega e_3\phi(u, 0). \end{aligned} \right. \quad (3.14)$$

For Einstein vacuum equation

$$\text{Ric}_{\mu\nu} = 0,$$

removing all ϕ -terms of ESE gives the iterative construction directly, so we omit the repetition of construction.

3.2 Curvature residues

Computing curvature usually involves second order derivatives, which would heavily reduce the numerical precision. Especially, $\lim_{v \rightarrow 0} R_{4A4B}(-1, v)$ can be infinite for short pulse perturbation and Ric_{44} is hard to compute in this way, even though it is theoretically zero in our construction. See [1] for the example of short pulse with infinite R_{4A4B} . We use intrinsic equations of Lorentzian manifold to compute the Ricci curvature:

$$\begin{aligned}\Omega^2 \text{Ric}_{33} &= -\Omega \nabla_3 (\Omega \text{tr } \underline{\chi}) - 4\Omega \underline{\omega} \Omega \text{tr } \underline{\chi} - \frac{1}{2} (\Omega \text{tr } \underline{\chi})^2 - |\Omega \widehat{\underline{\chi}}|^2, \\ \Omega \text{Ric}_{3A} &= \Omega \nabla_3 \zeta + \frac{3}{2} \Omega \text{tr } \underline{\chi} \zeta + \Omega \widehat{\underline{\chi}} \cdot \zeta + 2\nabla(\Omega \underline{\omega}) + \text{div}(\Omega \widehat{\underline{\chi}}) - \frac{1}{2} \nabla(\Omega \text{tr } \underline{\chi}) + \Omega \text{tr } \underline{\chi} \nabla \log \Omega, \\ \Omega \text{Ric}_{4A} &= -\Omega \nabla_4 \zeta - \frac{3}{2} \Omega \text{tr } \chi \zeta - \Omega \widehat{\chi} \cdot \zeta + 2\nabla(\Omega \omega) + \text{div}(\Omega \widehat{\chi}) - \frac{1}{2} \nabla(\Omega \text{tr } \chi) + \Omega \text{tr } \chi \nabla \log \Omega, \\ \Omega^2 \widehat{\text{Ric}}_{AB} &= \Omega \nabla_3 (\Omega \widehat{\chi}) + \frac{1}{2} \Omega \text{tr } \underline{\chi} \Omega \widehat{\chi} - \Omega^2 (\nabla \hat{\otimes} \eta + \eta \hat{\otimes} \eta) + \frac{1}{2} \Omega \text{tr } \chi \Omega \widehat{\underline{\chi}}, \\ \Omega^2 g^{AB} \text{Ric}_{AB} &= \Omega \nabla_3 (\Omega \text{tr } \chi) + \Omega \text{tr } \chi \Omega \text{tr } \underline{\chi} - 2\Omega^2 \text{div } \eta - 2\Omega^2 |\eta|^2 + 2\Omega^2 K, \\ \Omega^2 \text{Ric}_{34} &= 4\Omega \nabla_4 (\Omega \underline{\omega}) - \Omega \widehat{\chi} \cdot \Omega \widehat{\underline{\chi}} - \frac{1}{2} \Omega \text{tr } \chi \Omega \text{tr } \underline{\chi} + 4\Omega^2 \eta \cdot \underline{\eta} - \Omega \nabla_3 (\Omega \text{tr } \chi) + 2\Omega^2 \text{div } \eta.\end{aligned}$$

We only need the first order derivatives of the iterated quantities in this way.

3.3 Legendre–Gauss–Lobatto spectral elements

In the experiments, we will add non-spherical symmetric perturbation to the Einstein scalar-field or vacuum systems. To make the metric less regular, the perturbation is required to be of size $O(v^\delta)$. To deal with the non-smoothness, the characteristic coordinates are first replaced by coordinates in which the corner profiles are smoother. In particular, we use

$$\tau = -\log(-u), \quad s = \left(\frac{v}{v_{\max}} \right)^\delta, \quad u = -e^{-\tau}, \quad v = v_{\max} s^{1/\delta}.$$

Thus a profile proportional to v^δ is linear in s . This change of variables does not regularize the physical solution; it only represents its fractional behavior by a smooth function of the computational coordinate. Physical derivatives and integrals are recovered by

$$\partial_u = \frac{1}{-u} \partial_\tau, \quad \partial_v = \left(\frac{dv}{ds} \right)^{-1} \partial_s, \quad \int_0^v F(v') dv' = \int_0^s F(v(s')) \frac{dv}{ds'} ds'.$$

We divide each computational coordinate interval into elements. On an element of degree p , let $\{\xi_j\}_{j=0}^p$ be the Legendre–Gauss–Lobatto nodes, characterized by

$$(1 - \xi_j^2) P'_p(\xi_j) = 0,$$

including the two endpoints. A component F is represented by its Lagrange interpolant

$$I_p F(\xi) = \sum_{j=0}^p F(\xi_j) \ell_j(\xi).$$

The element differentiation and indefinite-integration matrices are

$$D_{ij} = \ell'_j(\xi_i), \quad Q_{ij} = \int_{-1}^{\xi_i} \ell_j(\xi) d\xi.$$

Consequently, interpolation of intermediate Runge–Kutta stages, differentiation, and cumulative integration are all obtained from the same element polynomial. Neighboring elements share their endpoint value, while the two derivative traces at an interior interface are combined into a single interface derivative. This composite construction permits local refinement in either characteristic direction without replacing the underlying transport hierarchy.

3.4 Tensor spherical-harmonic Galerkin discretization

The angular variables are represented on a quasi-uniform, pole-free point set on $\mathbb{S}^2$. Geometric vectors and tensors are stored as ambient Cartesian tensors and projected to the tangent bundle. This avoids the coordinate singularities of a polar chart. For a scalar field, we use the real spherical-harmonic expansion

$$f_L(\vartheta) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} f_{\ell m} Y_{\ell m}(\vartheta).$$

The coefficients are obtained by an overdetermined least-squares analysis of the nodal values, and angular derivatives are computed by differentiating the harmonic basis.

Vectors and symmetric two-tensors are not expanded componentwise as unrelated scalars. We use the electric and magnetic vector harmonics

$$Y_A^{E,\ell m} \sim \nabla_A Y_{\ell m}, \quad Y_A^{B,\ell m} \sim \epsilon_A^B \nabla_B Y_{\ell m},$$

and the trace, electric, and magnetic tensor harmonics

$$\gamma_{AB} Y_{\ell m}, \quad Y_{AB}^{E,\ell m} \sim (\nabla_A \nabla_B Y_{\ell m})^{\text{TF}}, \quad Y_{AB}^{B,\ell m} \sim \epsilon_{(A}^C \nabla_{B)} \nabla_C Y_{\ell m}.$$

These typed spaces preserve tangency, symmetry, and the tensorial transformation law. Trace-free quantities are projected with respect to the current section metric,

$$T_{AB}^{\text{TF}} = T_{AB} - \frac{1}{2}(\text{tr}_\gamma T)\gamma_{AB},$$

rather than with respect to a fixed background metric.

Let $\mathcal{V}_L$ denote the retained harmonic space and let $\mathcal{V}_W$, with $W > L$, be a work space used to form angular products. For a semidiscrete unknown $U_L \in \mathcal{V}_L$, every nonlinear right-hand side is defined by

$$\partial U_L = \Pi_L F(U_L),$$

where $F(U_L)$ is evaluated in the work space and Π_L is the appropriate scalar, vector, or tensor Galerkin projection. The retained/work separation reduces angular aliasing while keeping the evolved state in one declared finite-dimensional geometric space.

4 Results for Einstein Vacuum Equations

4.1 Crossed characteristic shears

We test genuinely nonspherical vacuum data on

$$\mathcal{Q} = \left\{ -1 < u < -\frac{1}{2}, \quad 0 < v < \frac{1}{2} \right\} \times \mathbb{S}^2.$$

At the corner set

$$\begin{aligned} g(-1, 0) &= \overset{\circ}{\gamma}, & \Omega(-1, 0) &= 1, & \Omega \operatorname{tr} \chi(-1, 0) &= 2, \\ \Omega \operatorname{tr} \underline{\chi}(-1, 0) &= -2, & \zeta(-1, 0) &= 0. \end{aligned}$$

We impose $\Omega(-1, v) = \Omega(u, 0) = 1$ and $b(u, 0) = 0.1\partial_\phi$. Writing $n \in \mathbb{S}^2$ and $P = I - n \otimes n$, define

$$\begin{aligned} A_1 &= \operatorname{diag}(-\tfrac{1}{2}, -\tfrac{1}{2}, 1), & A_2 &= \operatorname{diag}(1, -1, 0), & f_j(n) &= n^\top A_j n, \\ X_1 &= \frac{\sqrt{2}}{3} \operatorname{tf}_{\overset{\circ}{\gamma}} \nabla^2 f_1, & X_2 &= \frac{1}{2\sqrt{2}} \operatorname{tf}_{\overset{\circ}{\gamma}} \nabla^2 f_2, & \operatorname{tf}_{\overset{\circ}{\gamma}} \nabla^2 f_j &= 2PA_jP + f_jP. \end{aligned}$$

For a corner trace-free tensor X , let

$$\mathcal{T}_X[g] = \operatorname{tf}_g \left[\frac{1}{2} \left(X \overset{\circ}{\gamma}^{-1} g + (X \overset{\circ}{\gamma}^{-1} g)^\top \right) \right].$$

The free shears on the two characteristic faces are

$$\Omega \widehat{\chi}(-1, v) = v^{1/2} \mathcal{T}_{X_1}[g(-1, v)], \quad \Omega \widehat{\chi}(u, 0) = (u+1)^{1/2} \mathcal{T}_{X_2}[g(u, 0)]. \quad (4.1)$$

The remaining metric and expansion data are generated by the null constraints

$$\begin{aligned} \partial_v(\Omega \operatorname{tr} \chi) &= -\tfrac{1}{2}(\Omega \operatorname{tr} \chi)^2 - |\Omega \widehat{\chi}|_g^2, & \partial_v g &= (\Omega \operatorname{tr} \chi)g + 2\Omega \widehat{\chi}, \\ (\partial_u + b)(\Omega \operatorname{tr} \underline{\chi}) &= -\tfrac{1}{2}(\Omega \operatorname{tr} \underline{\chi})^2 - |\Omega \widehat{\underline{\chi}}|_g^2, & (\mathcal{L}_{\partial_u} + \mathcal{L}_b)g &= (\Omega \operatorname{tr} \underline{\chi})g + 2\Omega \widehat{\underline{\chi}}. \end{aligned} \quad (4.2)$$

The torsion is obtained from the corresponding $\nabla_4 \zeta$ and $\nabla_3 \zeta$ equations, while $\Omega \operatorname{tr} \chi$ on $\underline{H}_0$ is derived from its incoming null-structure equation. Hence the iteration receives constraint-generated characteristic data rather than independently sampled connection coefficients.

Using the evolved endpoint metrics and area forms, the immutable boundary artifact gives

$$\begin{aligned} \|\widehat{\chi}(-1, \tfrac{1}{2})\|_{L^2(S, g(-1, 1/2))} &= 2.727749, \\ \|\widehat{\underline{\chi}}(-\tfrac{1}{2}, 0)\|_{L^2(S, g(-1/2, 0))} &= 0.785641. \end{aligned} \quad (4.3)$$

Thus the perturbation is nontrivial on both initial null hypersurfaces.

The computation uses six degree-ten LGL elements in $t = \sqrt{2(u+1)}$ and $s = \sqrt{2v}$, giving 61×61 coordinate nodes, together with 300 sphere points, retained harmonic degree $L = 8$, work degree $W = 14$, and differentiation degree $D = 15$. All six characteristic slabs converge below the 10^{-8} Picard tolerance.

For every (u, v) define the six sphere residuals

$$\begin{aligned} r_{33} &= \|\Omega^2 \operatorname{Ric}_{33}\|_{L^2(S, g)}, & r_{34} &= \|\Omega^2 \operatorname{Ric}_{34}\|_{L^2(S, g)}, \\ r_{3A} &= \|\Omega \operatorname{Ric}_{3A}\|_{L^2(S, g)}, & r_{4A} &= \|\Omega \operatorname{Ric}_{4A}\|_{L^2(S, g)}, \\ r_{AB} &= \|\Omega^2 \widehat{\operatorname{Ric}}_{AB}\|_{L^2(S, g)}, & r_R &= \|\Omega^2 R\|_{L^2(S, g)}. \end{aligned}$$

Table 1 reports each $L_{u,v}^\infty L^2(S)$ residue separately. The protected interior removes the square-root endpoint elements and derivative halos; it is the accuracy statistic used for the nonsingular interior. The large open-grid value of r_{33} is confined to the unprotected $u = -1$ endpoint layer, where the physical $\partial_u = t^{-1}\partial_t$ derivative is singular. On the protected grid, 20 nodes lie between 10^{-3} and 10^{-2} and 124 between 10^{-4} and 10^{-3} .

Table 1: Componentwise $L_{u,v}^\infty L^2(S)$ Ricci residuals.

residue	open numerical grid	protected interior
r_{33}	1.371×10^3	1.689×10^{-3}
r_{34}	3.903	1.111×10^{-3}
r_{3A}	1.498×10^1	1.154×10^{-3}
r_{4A}	1.740	3.156×10^{-4}
r_{AB}	6.119	4.734×10^{-4}
r_R	9.185	1.319×10^{-3}

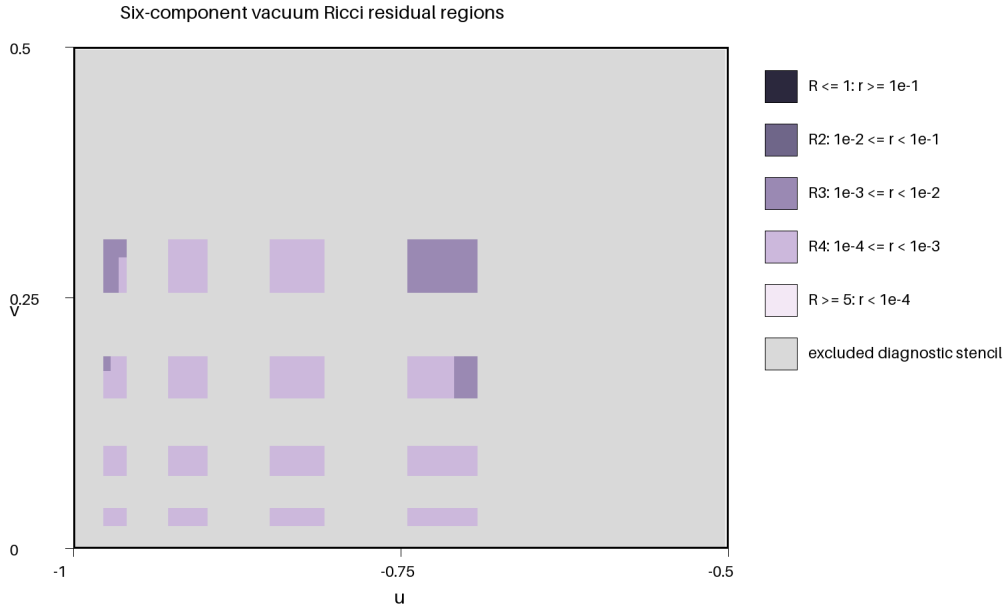


Figure 1: Spectrum of $r = r_{33} + r_{34} + r_{3A} + r_{4A} + r_{AB} + r_R$. Colors classify retained nodes by the residual thresholds 10^{-4} , 10^{-3} , 10^{-2} , and 10^{-1} . Gray cells are excluded by endpoint, interface, or derivative stencils.

5 Results for Einstein Scalar-field Equations

5.1 Regular exact scalar solutions

The Fisher–Janis–Newman–Winicour family supplies simultaneous exact metric and scalar controls. We use $\sigma = 1$ and $\nu \in \{0.99, 0.8, 0.5, 0.2\}$ on two, four, and eight degree-six elements. All twelve runs complete eight Picard sweeps. On the finest $\nu = 0.99$ grid, the scalar maximum absolute error is 1.14×10^{-14} and the sphere-metric maximum absolute error is 2.09×10^{-12} . The $\nu = 1$ control has identically vanishing scalar variables and recovers Schwarzschild mass $1/2$.

5.2 Nonspherical scalar refinement

Smooth real-harmonic lapse, shift, shear, and scalar data are completed by the characteristic Einstein–scalar constraints. The protected independent residual sum decreases along the coordinate sequence

$$28.18 \longrightarrow 0.429 \longrightarrow 0.0650$$

and along the angular sequence

$$0.529 \longrightarrow 0.429 \longrightarrow 0.411.$$

The associated closure envelopes decrease, and the five refinement cases end below a 10^{-6} Picard update. Two extreme stress probes are rejected before evolution because their incoming-scalar constraint radicand is negative. They are retained as failed positivity tests rather than omitted from the campaign.

5.3 Scalar pulse and trapped-section sign audit

For the pulse test we take $\kappa = 0.25$, $\delta = 0.1$, scalar amplitude 2, profile scale 0.01, and quadrupolar shear amplitude 0.1. A twelve-element base domain is prolonged by one complete tau element. The standard angular band $(L, W, N) = (5, 10, 170)$ and control band $(7, 14, 300)$ independently evaluate the two expansion suprema on 1000 sphere points.

The base update reaches 3.58×10^{-6} in the standard band and 1.32×10^{-5} in the control. The continued updates remain 1.76×10^{-3} and 9.90×10^{-3} . Their protected negative-sign masks contain 18 and 16 cells, with all 16 control cells contained in the standard mask. However, the protected independent Einstein–scalar residual sums are 9.64×10^3 and 6.98×10^3 . Thus the sign observation is neither a numerical candidate nor a certificate: the Picard, residual, and coordinate-refinement gates are not met.

6 Conclusion

The weighted first-order variables expose a triangular Picard hierarchy for the characteristic vacuum and Einstein–scalar equations. Immutable face data, trace-free angular projection, spectral elements, and independent four-metric audits turn that hierarchy into a testable numerical method. Exact regular solutions and the crossed-pulse slab evolution provide the strongest validations. Near a singular radius, in a degenerate static horizon chart, and in the continued scalar pulse, the diagnostics also show where a completed computation does not support an accuracy or existence claim. Those distinctions between execution, candidate status, and certificate status are part of the numerical result.

References

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